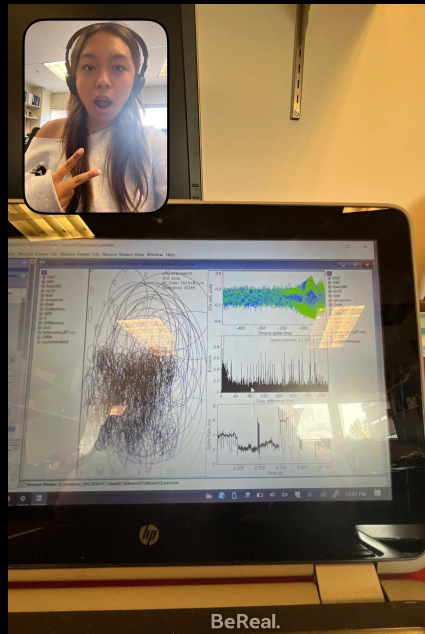


BRIGHT-UP

Biomedical Research Internship for the Growth of
High-school Trainees from Underrepresented Populations



Tate Suarez

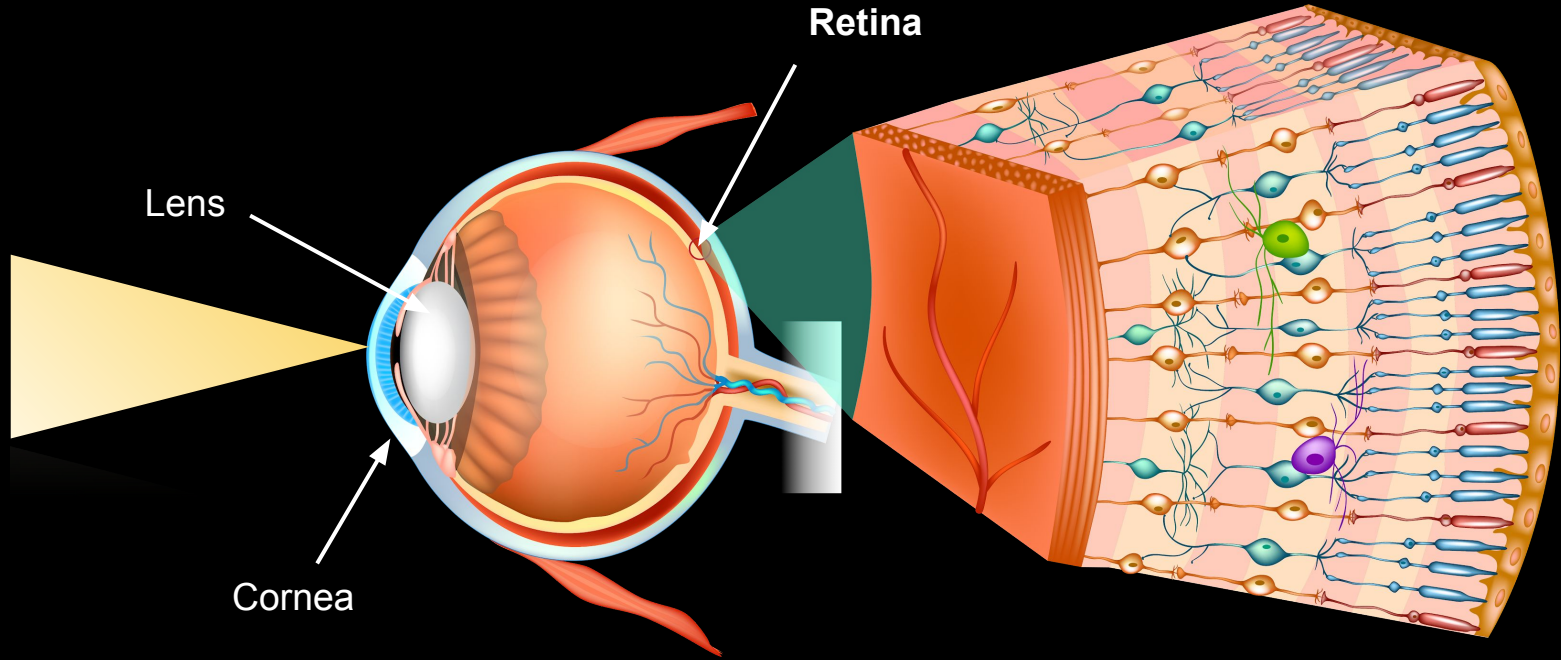
Senior, 12th - Raisbeck Aviation
High School

Rieke Lab

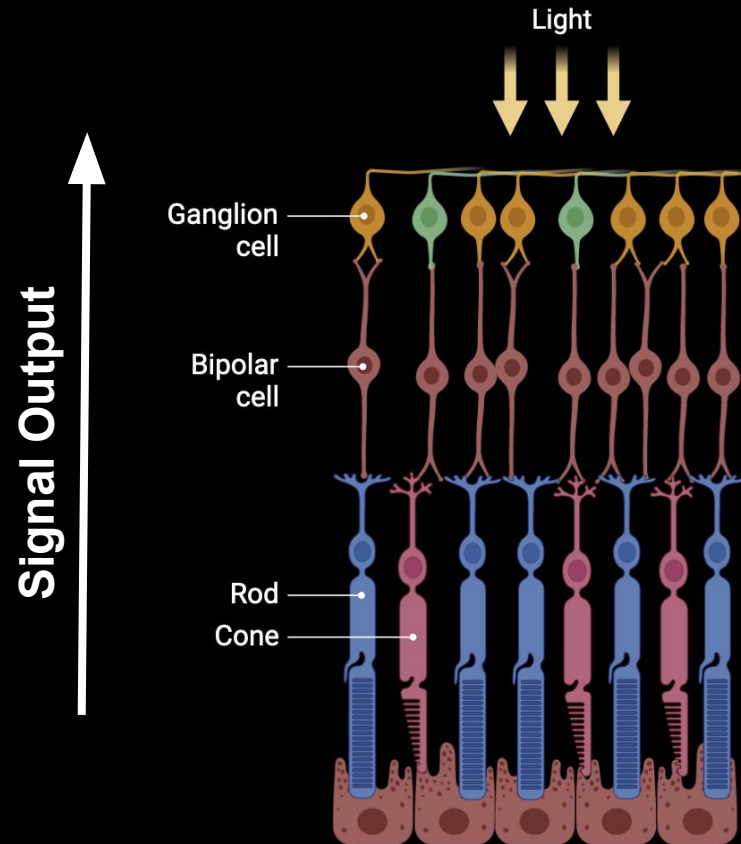


The Rieke Lab focuses on how the retina converts light into electrical signals
your optic nerve sends to your brain

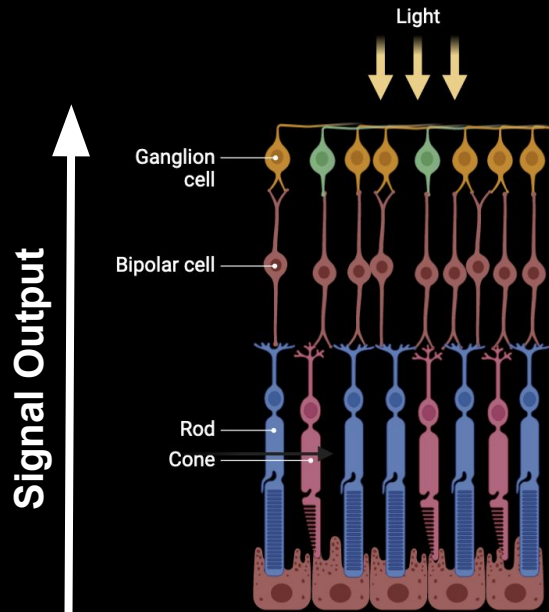
How does the eye perceive visual stimuli?



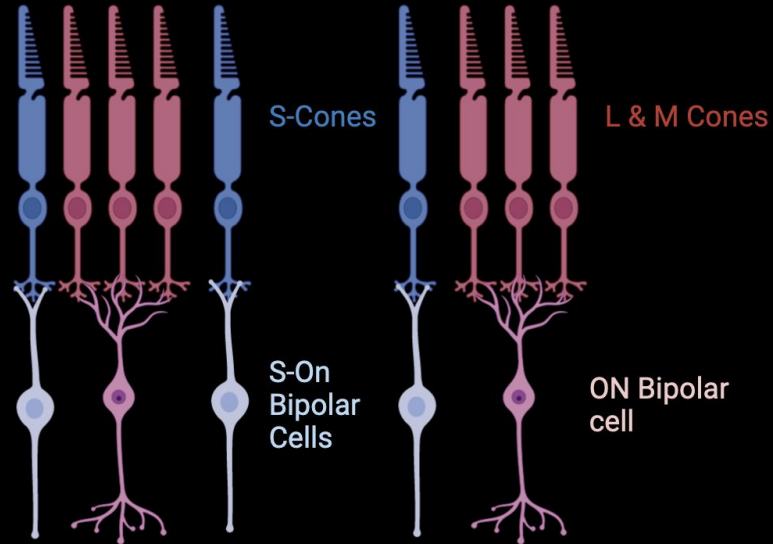
The mechanics of retinal circuitry



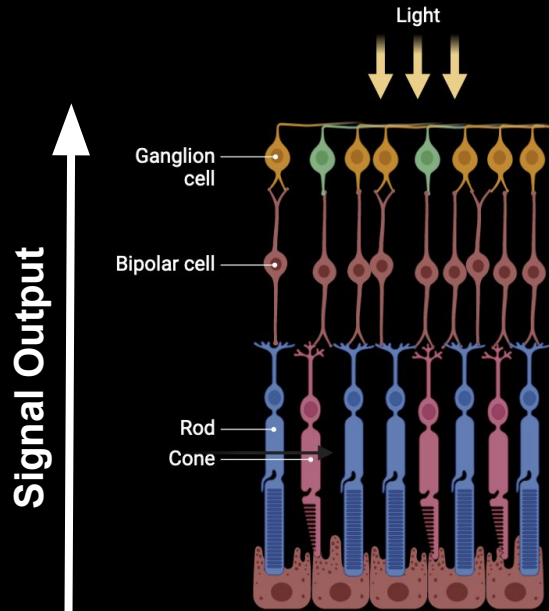
Photoreceptors



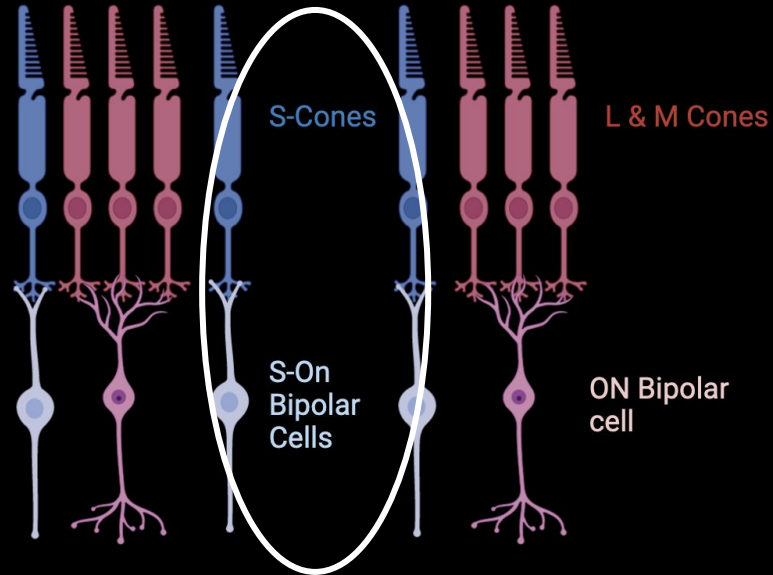
Cones: Responsible for what we see in normal lighting



Photoreceptors



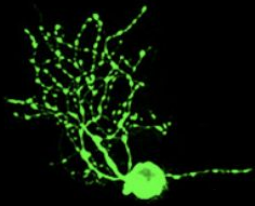
Cones: Responsible for what we see in normal lighting



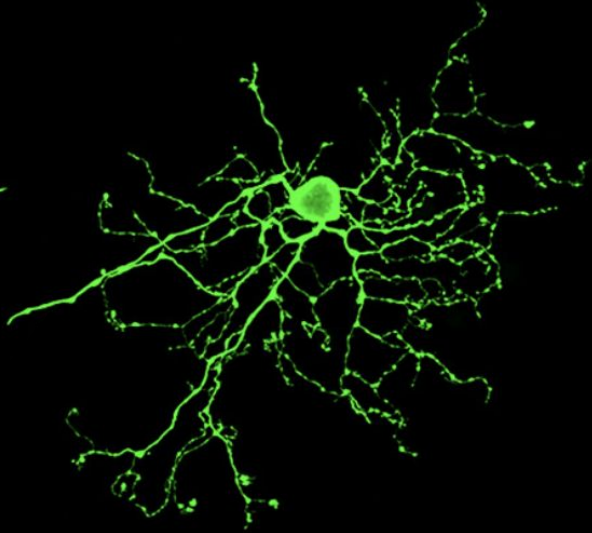
Ganglion cell types



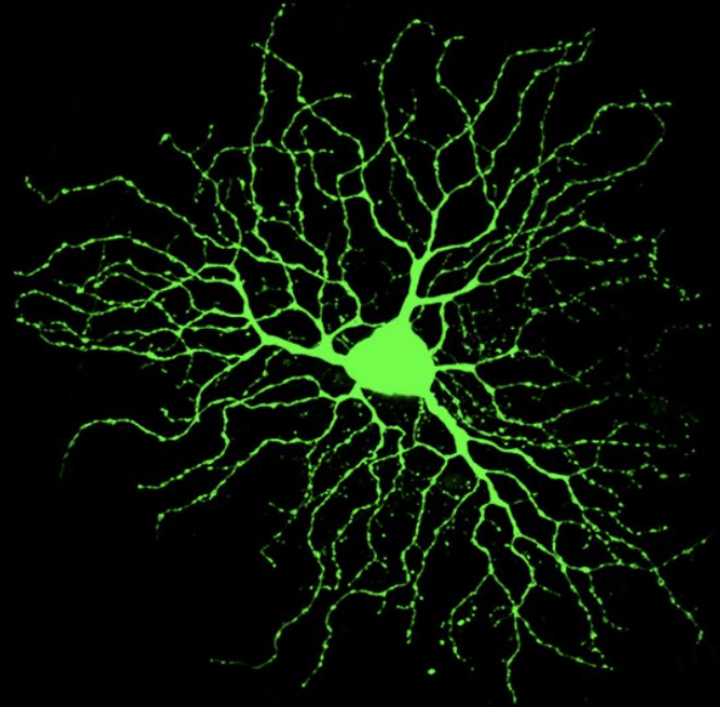
Midget



Small bistratified



Parasol

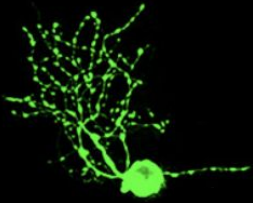


3 ganglion cell types make up 70% of
all ganglion cells

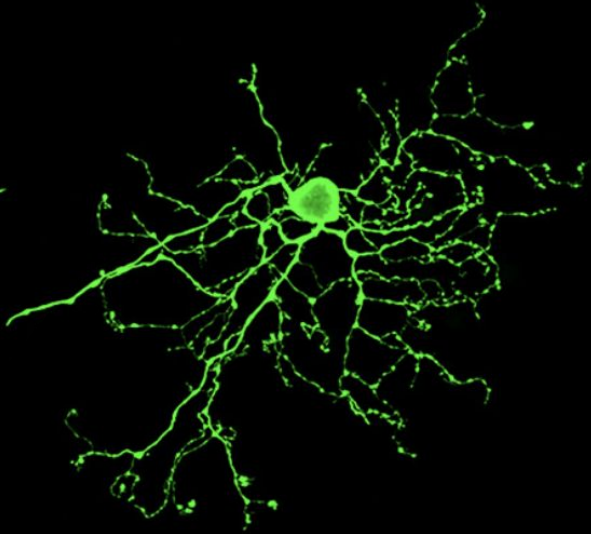
Ganglion cell types



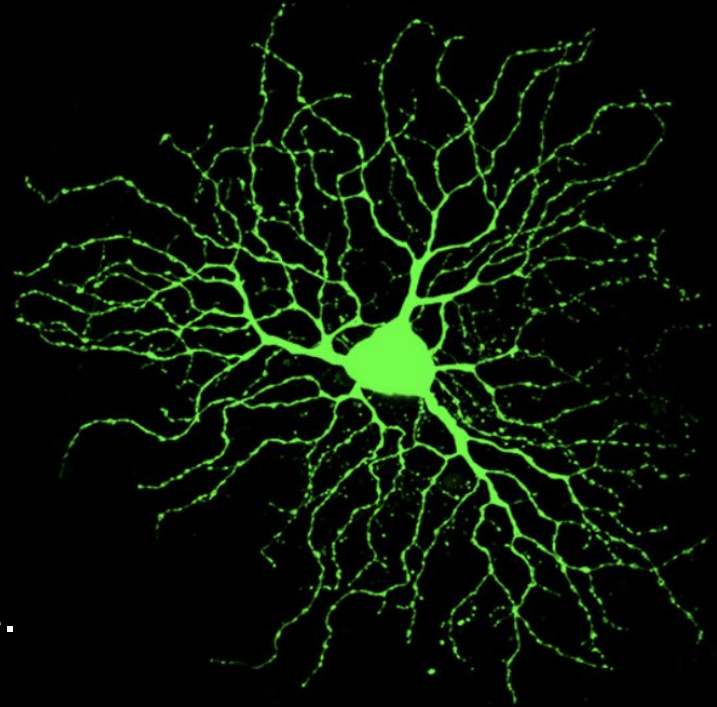
Midget



Small bistratified



Parasol

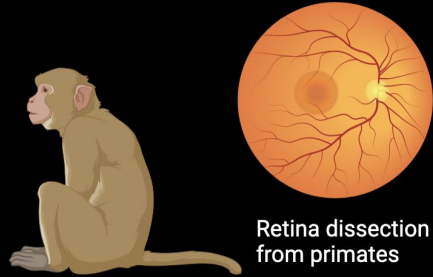


30% are known as rare ganglion cell types.

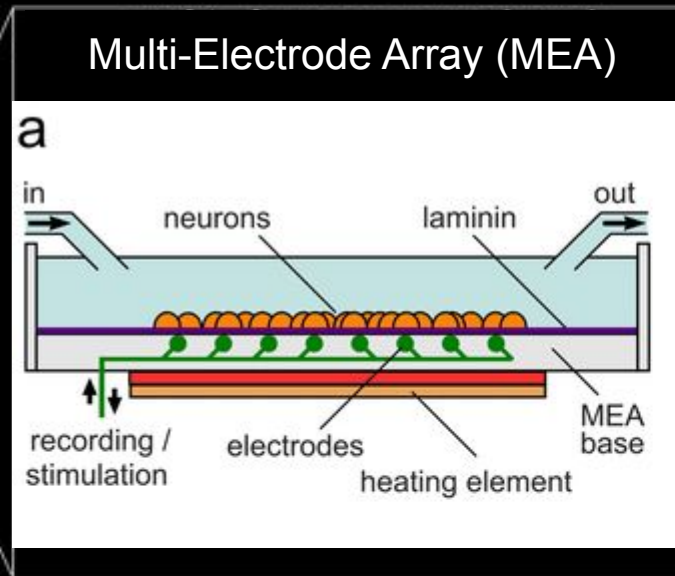
- Understudied

How many retinal ganglion cell types receive S-cone input?

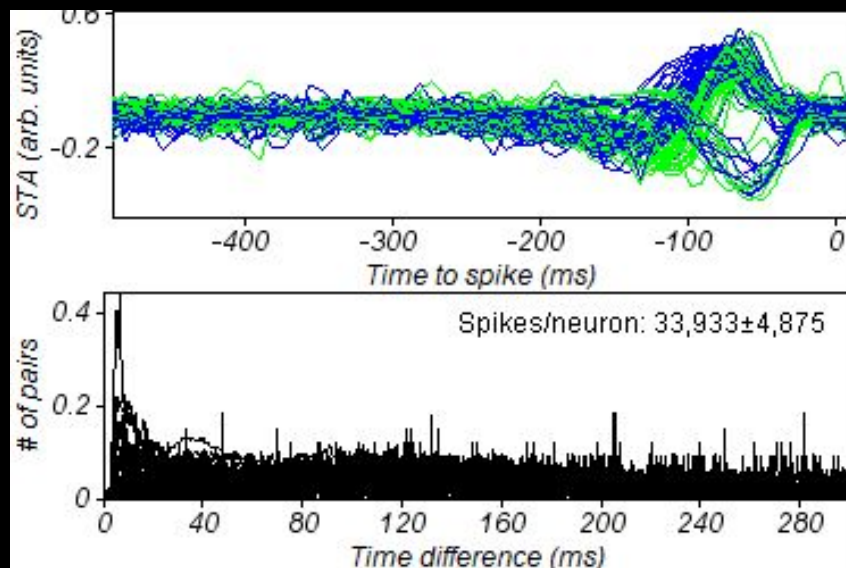
Multi-Electrode Array (MEA)



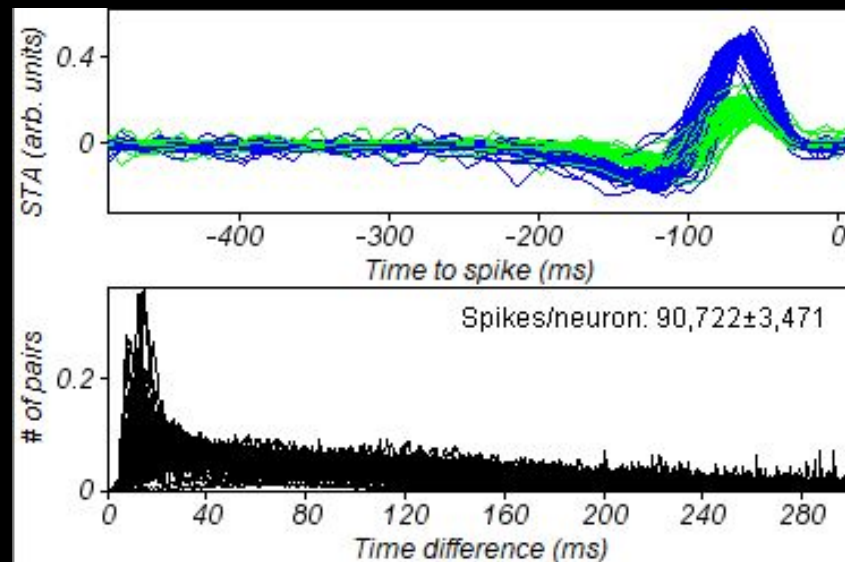
Retina dissection
from primates



S-Cone (Non-SBC) Cells

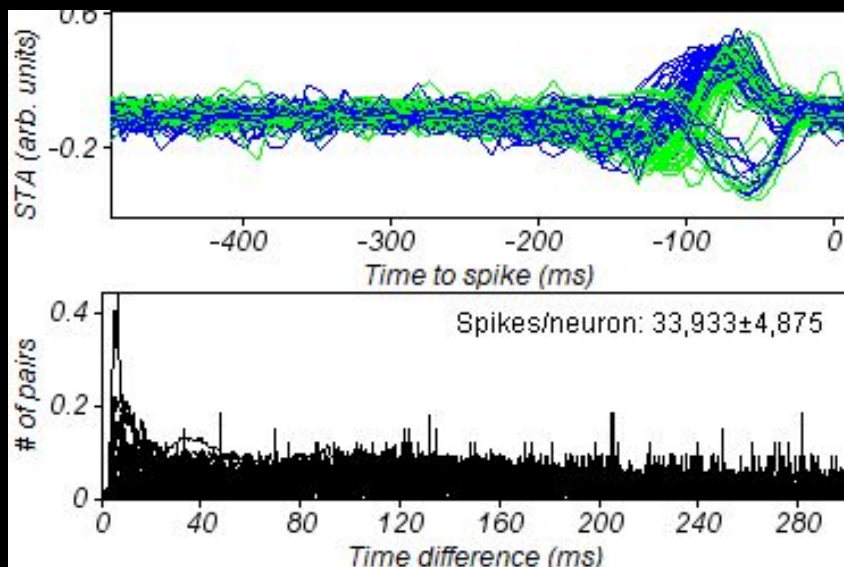


SBC (Small Bistratified Cells)



- S Cone Input
- L & M Cone Input

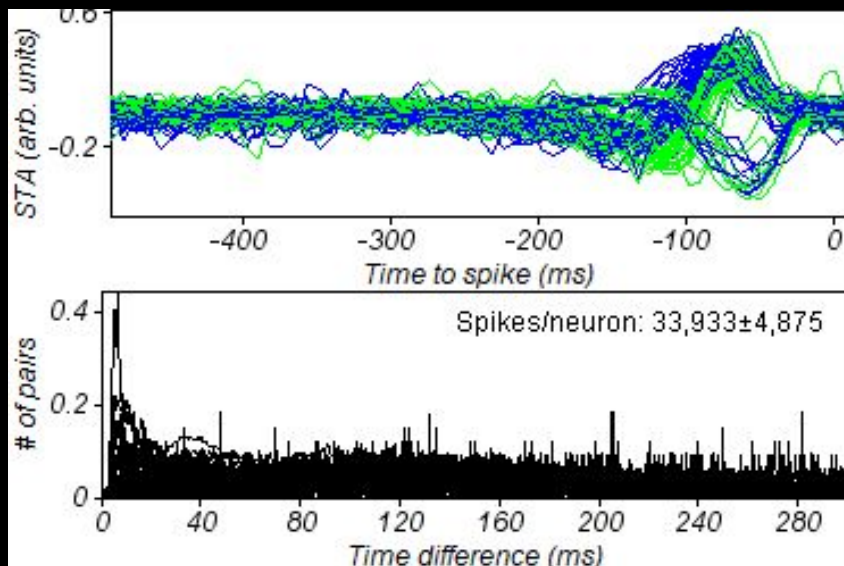
S-Cone (Non-SBC) Cells



There are multiple temporal profiles indicating the diversity of those non-SBC cells.

-  S Cone Input
-  L & M Cone Input

S-Cone (Non-SBC) Cells



There are multiple temporal profiles indicating the diversity of those non-SBC cells.

WHAT WOULD BE NEXT?

Cell typing non-SBCs

- On/Off Responses
- L & M Cone sensitivity

Analyze their response properties to a library of both natural and artificial stimuli

-  S Cone Input
-  L & M Cone Input

Thank You!



- Thank you Claudia and Fred for creating the Program!
- Lizbeth de la Cruz from Making Connections Program
- Thank you to the Rieke/Manookin Lab
 - Fred Rieke
 - Mike Manookin
 - Chris Chen
 - Arthur Hong
 - Vyom Raval
 - Shane Lind-Boniec
 - Srinidhi Naidu
 - Todd Appleby
 - Nori Ingram

